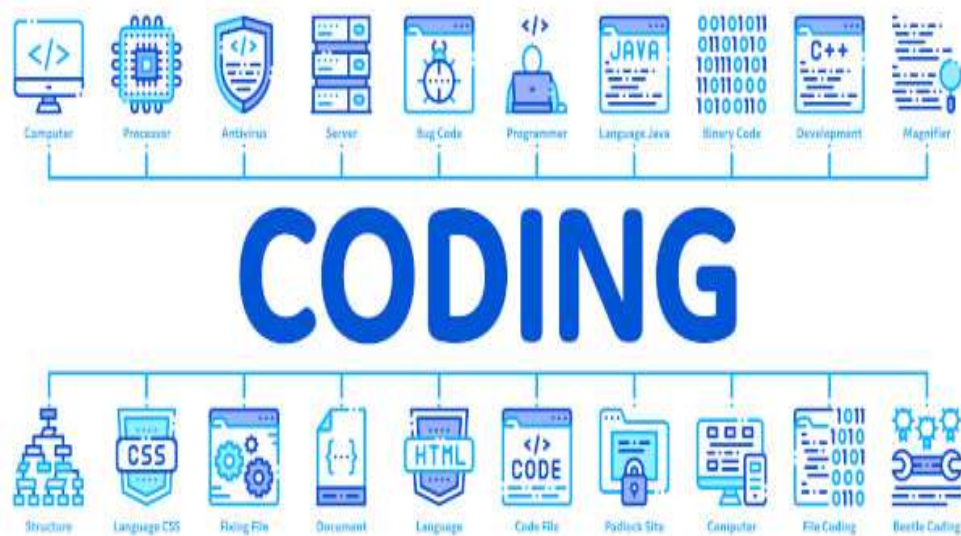


MODULE : OPEN SOURCE CODING

MODULE CODE : OPSC6311

POE – PART 1



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PLANNING AND DESIGN

Introduction

Based on the research current budgeting apps like WalletCorner, Goodbudget, and PocketGuard shows that contemporary budgeting tools integrate automation, organization, and user involvement to enhance efficient financial management. Every app presents distinct advantages, such as gamification (WalletCorner), envelope-style budgeting (Goodbudget), and automated tracking of expenses (PocketGuard).

This design aims to create a Student Budget App specifically designed for students in South Africa. Numerous students encounter financial difficulties like coping with restricted allowances, managing transportation expenses, purchasing data, and balancing school and personal costs. Consequently, the suggested app seeks to offer an easy, enjoyable, and efficient way to handle finances successfully. This design will integrate the finest characteristics found in the research while modifying them to meet student requirements and the South African context.

Overview

App Name: EduBudget

App Icon Design:



Figure 1: A wallet topped with a graduation cap. Colours: Green (finance/development), Blue (confidence), White (clarity)
This signifies student living and money handling.(design of the app icon)

Description of Innovative Features:

Gamification System:

Users receive points, badges, and rewards for:

- Remaining within budget
- Recording daily expenditures
- Reaching financial objectives

Envelope Budgeting System:

Users can allocate money into categories such as:

- Groceries
- Transport
- Airtime/Data
- Entertainment
- Savings

“Safe to Spend” Feature (Inspired by PocketGuard)

shows the student how much money can be spent after expenses.

Manual or Optional Auto Tracking

South African banks may not fully integrate the supports of the app

- Manual entry : student have to physically do their transactions on the app.
- Optional linking : students wouldn't need to do anything the app would do it for them each month to their preferred layout.

Student-Focused Categories:

categories relevant to SA students:

- NSFAS/Allowance
- Rent/Res
- Taxi fare
- Data bundles

Progress Dashboard:

Shows:

- Spending progress
- Monthly budget usage
- Overspending alerts

Detailed Requirements

Functional Requirements:

The app must:

- Allow users to **register and log in**
- Allow users to **create and manage expense categories**
- Allow users to **add expense entries** (amount, date, category, description)
- Allow users to **attach receipts**
- Allow users to **set a monthly budget and category limits**
- Allow users to **view expenses over a selected period**
- Allow users to **view total spending per category**
- Store all data in a **local database**
- Display **graphs of spending trends**
- Provide a **dashboard showing budget progress**
- Highlight **overspending categories**
- Include **gamification elements**

Non-Functional Requirements:

- The app must be **user-friendly and easy to navigate**
- Must work on **Android devices**
- Must support **offline functionality**
- Must load quickly and respond efficiently
- Must ensure **data privacy and security**

Other features:

- Daily spending reminders
- “No Spend Day” challenge
- Savings goal tracker
- Dark mode (for accessibility)
- Currency set to **South African Rand (R)**

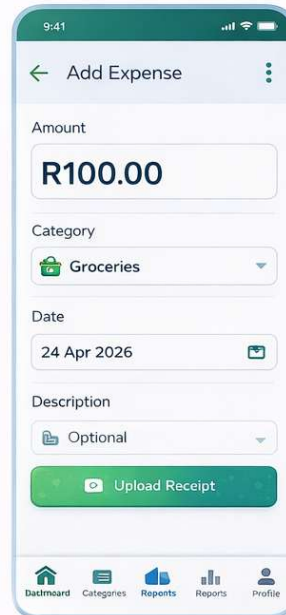
User Interface Design:



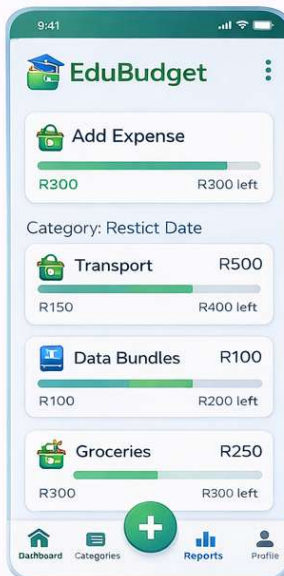
Screen 1: Login/Register Screen



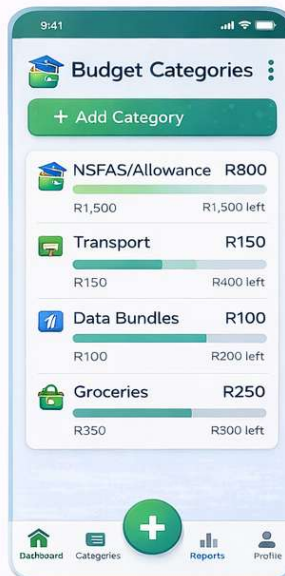
Screen 2: Dashboard



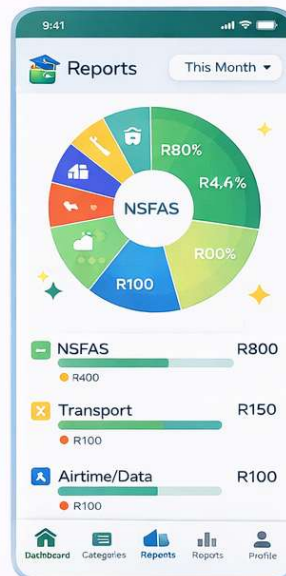
Screen 3: Add Expense Screen



Screen 1: Add Expense Screen



Screen 4: Budget Categories



Screen 5: Reports/Graphs

Screen 1 :Login/Register Screen

Purpose:

Allow users to create an account or log in.

Features:

- Username & password fields
- Login button
- Register option

Screen 2: Dashboard

Purpose:

Provide an overview of financial status.

Features:

- Total balance
- “Safe to Spend” amount
- Budget progress bar
- Quick add expense button

Screen 3: Add Expense Screen

Purpose:

Allow users to record expenses.

Features:

- Amount input
- Category dropdown
- Date selection
- Description
- Upload receipt

Screen 4: Budget Categories

Purpose:

Manage spending categories.

Features:

- Add/edit/delete categories
- Set limits per category

Screen 5: Reports/Graphs

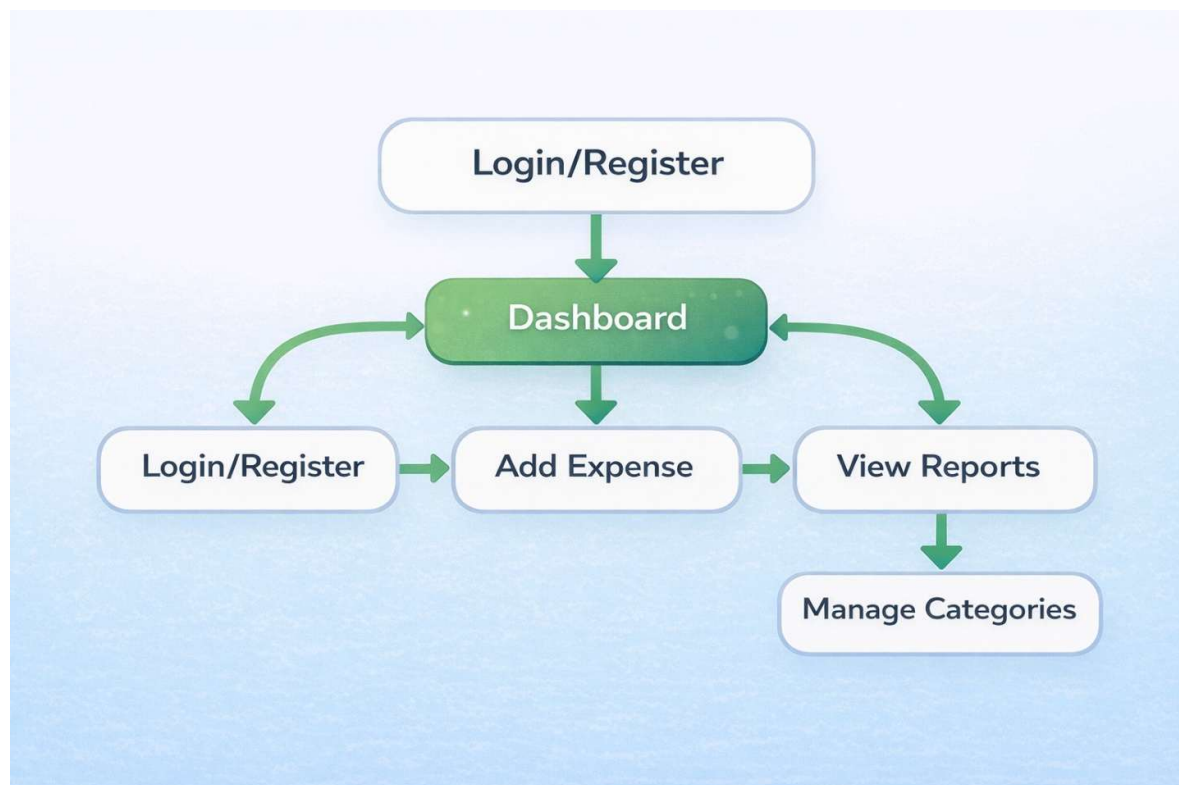
Purpose:

Visualize spending trends.

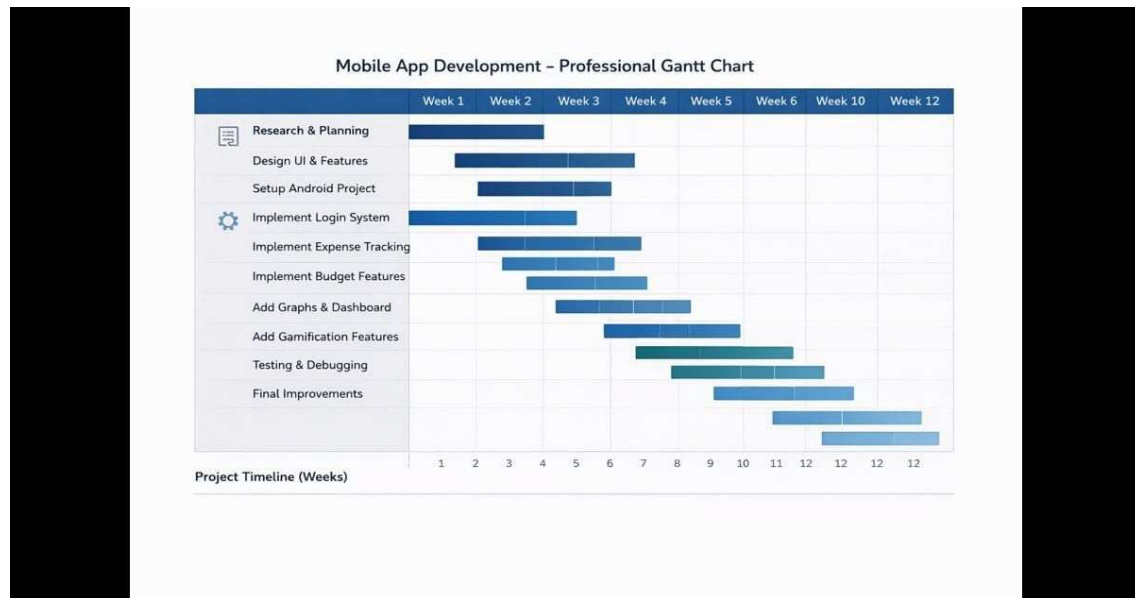
Features:

- Pie chart (category spending)
- Bar graph (daily/monthly spending)

Navigation Flow Diagram



Project Plan (Gantt Chart)



Conclusion

In conclusion the suggested Student Budget App, EduBudget, aims to tackle the financial difficulties encountered by students in South Africa. The app seeks to enhance budgeting by integrating gamification, organizing budgeting, and streamlined financial tracking.

Taking from what we learnt about the advantages of WalletCorner, Goodbudget, and PocketGuard, this design combines the best features into one platform designed for student requirements. The addition of localized categories, offline capabilities, and an intuitive design makes the app useful and pertinent.

This design establishes a solid basis for creating a practical and efficient budgeting app.

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